

Kacem Mathlouthi

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Education

National Institute of Applied Science and Technology (INSAT)

Sept 2022 – May 2027

Engineering Degree in Software Engineering

Experience

Software Engineer

On Site

Clusterlab [🔗](#) – Conversational Voice AI Agents

Jun 2025 – Present

- Built **Web Call** feature end-to-end across full stack, delivering browser-based voice calls that eliminated telephony providers.
- Developed and published **webcall-widget** [🔗](#) to **npm**, enabling seamless third-party voice agent integration.
- Engineered high-performance Call Recorder microservice in **Rust**, handling concurrent recording sessions.
- Reduced average latency by **250ms** through audio mixing and noise cancellation optimization, significantly improving UX.
- Built **LLM-powered** post-call analysis pipeline to automatically extract actionable insights and conversation outcomes.

Google Summer of Code Developer

Remote

NRNB (National Resource for Network Biology)

May 2025 – Sep 2025

- Collaborated with the VCell team to develop and deploy **VCell-AI** [🔗](#), an AI agent platform with **function-calling** capabilities, featuring a **Next.js** frontend and **FastAPI** backend, enabling researchers to query, explore, and create biological models.
- Integrated **RAG** system using **Qdrant** to provide contextual information about VCell software, tutorials, and guidelines.
- Enabled AI agent to analyze diagrams and VCML files using multimodal capabilities, supporting biological model interpretation.
- Implemented **LLM observability** via **Langfuse** to track agent interactions, monitor performance, and improve reliability.

Machine Learning Engineering Intern

Tunis, Tunisia

Orange

Aug 2024 – Oct 2024

- Developed lightweight **CNN model** for plant disease classification optimized for **edge devices** using **knowledge distillation**.
- Built **AI agentic workflow** for automated report generation, streamlining data collection, analysis, and research extraction.
- Achieved **4th place** in Orange Summer Challenge, pitching the solution to both technical and non-technical stakeholders.

Publications

Revisiting Generalization Measures Beyond IID: An Empirical Study under Distributional Shift

Under review at International Conference on Machine Learning (ICML) 2026

[arXiv:2602.01718](#) [🔗](#)

- Conducted large-scale empirical study benchmarking 40+ generalization measures, extending evaluation beyond IID settings to assess robustness under diverse distribution shifts.

Projects

Metis AI: FastAPI, React, Celery, Redis, PostgreSQL, GitHub App, GCP

[Live Demo](#) [🔗](#)

- Built and deployed a full-stack **GitHub App** that automates pull request reviews using **AI agents** in a sandboxed environment.
- Designed asynchronous processing with **Celery + Redis** to handle webhook-driven review workflows and background issue-to-PR coding tasks.

Achievements and Recognitions

1st place at Orange x Hexabot Conversational AI Hackathon

[devpost/x2x-Submission](#) [🔗](#)

- Built extensions for Hexabot visual editor, enabling multimodal capabilities like text-to-speech, speech-to-text, and vision.

2nd place at IEEE TSYP 12 CS Technical Challenge

[Smartshield-INSAT](#) [🔗](#)

- Earned 2nd place at the Tunisian Students and Young Professionals Computer Science Challenge by creating an AI-powered platform for automated cybersecurity threat detection and response, Smartshield.

2nd place at Artificial Intelligence National Summit (AINS 2.0)

[AINS-ML.Guide](#) [🔗](#)

- Implemented a Multi-LLM Agent System to help engineers define, evaluate, and solve machine learning problems.

Technologies

Languages: Python, Typescript, Go, Rust, SQL

Tools & Technologies: Git, Linux, Docker, NGINX, Azure, GitHub/GitLab, MongoDB, PostgreSQL, Vector Databases, React, Next.js, FastAPI, Flask, Express.js, LangChain, RAG Systems, LLM Workflows, Langfuse